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United States Patent	12410009
Kind Code	B2
Date of Patent	September 09, 2025
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Automated storage and retrieval system

Abstract

There is provided an automated storage and retrieval system including: a rack storing a plurality of bins for containing items; a transportation robot transporting the bin in the rack; and a transportation elevator which moves in the rack and can transport the transportation robot. The rack includes a plurality of floors each of which stores each of the plurality of bins, and allows the transportation robot to run along a surface thereof, and the rack includes a plurality of support column modules forming a passage of the transportation elevator. Each of the support column modules extends between at least mutually neighboring floors, and one support column module is configured to be able to be coupled to the another support column module.

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Appl. No.: 18/165375

Filed: February 07, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230278794 A1	Sep. 07, 2023

Related U.S. Application Data

us-provisional-application US 63315563 20220302
us-provisional-application US 63482805 20230202

Publication Classification

Int. Cl.: **B65G1/04** (20060101); **B25J9/16** (20060101); **B25J13/00** (20060101); **B65G1/06** (20060101); **B65G1/10** (20060101); **B65G1/127** (20060101); **B65G1/137** (20060101); **B25J5/00** (20060101)

U.S. Cl.:

CPC **B65G1/0492** (20130101); **B25J9/161** (20130101); **B25J9/1676** (20130101); **B25J13/006** (20130101); **B65G1/0471** (20130101); **B65G1/0478** (20130101); **B65G1/06** (20130101); **B65G1/065** (20130101); **B65G1/10** (20130101); **B65G1/127** (20130101); **B65G1/1371** (20130101); **B65G1/1373** (20130101); **B25J5/007** (20130101); **B65G2201/0235** (20130101); **B65G2201/0258** (20130101); **B65G2203/0233** (20130101); **B65G2203/0283** (20130101); **B65G2811/0678** (20130101)

Field of Classification Search

CPC: B65G (1/0464); B65G (1/0492); B65G (1/10); B65G (1/0471); B65G (1/127); B65G (1/04); B65G (1/065); B65G (2201/0235); B65G (2201/0414); B65G (2201/0492); B65G (2203/0233); B65G (2203/0283); B65G (2811/0678)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of priority to U.S. Provisional Application No. 63/315,563 filed on Mar. 2, 2022, and U.S. Provisional Application No. 63,482/805 filed on Feb. 2, 2023, the contents of each of which are hereby incorporated by reference in their entirety

TECHNICAL FIELD

(1) The present disclosure generally relates to a warehouse system such as an Automated Storage and Retrieval System (an ASRS or an AS/RS).

BACKGROUND

(2) As Electronic Commerce (EC) markets expand, improvement of item storage efficiency and item picking work efficiency for logistics has become a great problem. In order to resolve this problem, for example, WO2018/189110 A1 discloses a warehouse. This warehouse is partitioned into a storage area **10** and an order preparation area **11**. Rack **100** storing trays **102** for containing storage target products are arranged in the storage area **10**. On the other hand, order preparation stations **12** for operators **13** to process orders are arranged in the order preparation area **11**.

(3) To process an order, an automated device **103** transports the tray **102** containing an order target product from the rack **100** to the order preparation station **12**. Subsequently, in the order preparation area **11**, the tray **102** is transported to in front of the operator **13** by, for example, another operator other than the operator **13**, or another system (e.g., conveyer). The operator **13** picks the product from the tray **102**, and sorts the picked product as a transportation package **14**. As EC markets expand, it is demanded to make such work more efficient. Furthermore, a structure enabling flexible design of a form of an automated storage and retrieval system at various places at which the automated storage and retrieval system is installed is demanded.

(4) The present disclosure has been made in light of the above problem, and an object of the present disclosure is to provide an automated storage and retrieval system which enable flexible design of a form thereof.

SUMMARY

(5) In order to achieve the above object, according to one aspect of the present disclosure, an automated storage and retrieval system is provided, and the automated storage and retrieval system includes: a rack storing a plurality of bins for containing items; a transportation robot transporting the bin in the rack; and a transportation elevator which moves in the rack and can transport the transportation robot. The rack includes a plurality of floors each of which stores each of the plurality of bins, and allows the transportation robot to run along a surface thereof. Furthermore, the rack includes a plurality of support column modules forming a passage of the transportation elevator. Each of the support column modules extends between at least the mutually neighboring floors. One support column module is configured to be able to be coupled to another support column module.

(6) Each of the support column modules includes an engagement member guiding the transportation elevator.

(7) The transportation elevator includes at least one driven part to be rotationally driven, and the engagement member is configured to be able to be engaged with the driven part of the transportation elevator.

(8) The driven part of the transportation elevator is a sprocket, and the engagement member is a roller chain.

- (9) Each of the support column modules is configured to be able to support the floor adjacent to the passage of the transportation elevator.
 - (10) The engagement member of each of the support column modules is configured to be located in a space of the passage of the transportation elevator. Each of the support column modules is configured to be able to support the floor outside the space of the passage of the transportation elevator.
 - (11) Each of the support column modules includes an insertion plug, an accommodation plug, and a main body extending between the insertion plug and the accommodation plug. The insertion plug and the accommodation plug are configured to be able to be coupled to each other.
 - (12) The engagement member of the one support column module is configured to be able to be coupled to the engagement member of the another support column module.
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Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view schematically illustrating an external appearance of an automated storage and retrieval system 1 according to an embodiment of the present disclosure;
- (2) FIG. 2 is a top view schematically illustrating an example of a structure of floors 11 of a rack 10 of the automated storage and retrieval system 1 according to the present embodiment;
- (3) FIG. 3 is a perspective view schematically illustrating an example of a structure of the rack 10 of the automated storage and retrieval system 1 according to the embodiment;
- (4) FIG. 4 schematically illustrates an example of a structure of a storage bin 20 of the automated storage and retrieval system 1 according to the embodiment;
- (5) FIG. 5 is a perspective view schematically illustrating an external appearance of a transportation robot 30 of the automated storage and retrieval system 1 according to the embodiment;
- (6) FIG. 6 is a side view illustrating a state where the transportation robot 30 of the automated storage and retrieval system 1 according to the embodiment lifts the storage bin 20;
- (7) FIG. 7 is a functional block diagram schematically illustrating a configuration of a management server 50;
- (8) FIG. 8 is a functional block diagram schematically illustrating a configuration of the transportation robot 30;
- (9) FIG. 9 is a side view of the rack 10 schematically illustrating an example of a configuration of a picking station 40 of the automated storage and retrieval system 1 according to the embodiment;
- (10) FIG. 10 is a plan view of the rack 10 schematically illustrating an example of a configuration of the picking station 40 of the automated storage and retrieval system 1 according to the embodiment;
- (11) FIG. 11 is a perspective view illustrating a schematic configuration of a rack module 2 included in the rack 10;
- (12) FIGS. 12A and 12B are perspective views of a support column module 70 included in the rack module 2;
- (13) FIG. 13A is a plan view of the support column module 70;
- (14) FIG. 13B is a right side view of the support column module 70;
- (15) FIG. 13C is a left side view of the support column module 70;
- (16) FIG. 13D is a top view of the support column module 70;
- (17) FIG. 13E is a bottom view of the support column module 70;
- (18) FIG. 13F is a back view of the support column module 70;
- (19) FIG. 13G is an enlarged view of FIG. 13A;
- (20) FIG. 14 is a perspective view of a floor module 80 included in the rack modules 2;

- (21) FIG. 15 is a view illustrating an enlarged vicinity of a corner 83 of the floor module 80;
- (22) FIG. 16 is a partial exploded perspective view of the rack module 2;
- (23) FIG. 17 is an exploded perspective view of the rack module 2;
- (24) FIGS. 18A and 18B are perspective views of a support column end module 77 included in the rack 10;
- (25) FIGS. 19A and 19B are perspective views illustrating the plurality of rack modules 2 connected in a horizontal direction and a vertical direction;
- (26) FIG. 20 is a perspective view of a support column module according to a modified example;
- (27) FIG. 21 is a perspective view illustrating an example of a hook included in the support column module according to the modified example;
- (28) FIG. 22 is a side view of a support column module for forming a passage of a transportation elevator;
- (29) FIG. 23 is a perspective view of the support column module illustrated in FIG. 22;
- (30) FIG. 24 is a perspective view of the support column module illustrated in FIG. 22;
- (31) FIG. 25 is an exploded perspective view of the support column module illustrated in FIG. 23;
- (32) FIG. 26 is a view for illustrating a configuration for coupling roller chains of the support column module illustrated in FIG. 23; and
- (33) FIG. 27 is a perspective view of a transportation elevator and a part of a passage of the transportation elevator in the rack.

DESCRIPTION OF EMBODIMENTS

(34) Hereinafter, an embodiment of the present disclosure will be described with reference to the accompanying drawings. In all drawings, identical reference numerals are used to refer to identical or similar components. The following embodiment does not intend to limit the disclosure recited in the claims. Examples and features of the disclosed principal will be described in this description, yet can be changed and modified without departing from the idea and the scope of the disclosed embodiment. Furthermore, specific features, structures, or characteristics can be combined by an arbitrary appropriate method in one or more embodiments. The following detailed description is taken into account only as exemplary description, and the true range and idea are intended to be indicated by the claims.

(35) FIG. 1 is a perspective view schematically illustrating an external appearance of an automated storage and retrieval system 1 according to an embodiment of the present disclosure. The automated storage and retrieval system 1 is constructed in, for example, a building 100. The building 100 includes a floor surface 101 extending along a horizontal plane, a plurality of sidewalls 102 vertically standing from the floor surface 101, and a ceiling (not illustrated) supported by the plurality of sidewalls 102. For example, the four sidewalls 102 surrounding four sides, and the ceiling extending in parallel to the floor surface 101 establish an internal space between the floor surface 101, and the sidewalls 102 and the ceiling. The automated storage and retrieval system 1 is constructed in this internal space. Note that part of the sidewalls 102 are cut out for ease of description in FIG. 1. In the building 100, an x axis and a y axis extending in a direction perpendicular to each other on the horizontal plane, and a z axis extending in a vertical direction and perpendicular to the x axis and the y axis are defined.

(36) The automated storage and retrieval system 1 is an automated storage and retrieval system which can automate a series of work from warehousing and storage to delivery of items including products and the like based on centralized management. The automated storage and retrieval system 1 according to the present embodiment includes a rack 10 arranged on the floor surface 101, a plurality of storage bins 20 which are containers stored in the rack 10, a plurality of transportation robots 30 for transporting the storage bins 20, one or more picking stations 40 for picking items including products and the like contained in the storage bins 20, and a management server 50 for managing the series of work of the automated storage and retrieval system 1. At the picking station 40, a human operator 60 performs picking work of picking an item from the storage bin 20.

(37) The rack **10** includes a plurality of floors **11**, each of floors **11** defining a surface which extends in parallel to each other along each xy plane, and a plurality of support columns **12** supporting the plurality of floors **11**. In the present embodiment, the floors **11** of a first floor to a ninth floor vertically standing in an x axis direction from the floor surface **101** are formed. A total height of the rack **10** from the floor surface **101** in the x axis direction can be set according to a height in the z axis direction of the ceiling of the building **100**. Note that part of the floors **11** of the eighth floor and the ninth floor is omitted in FIG. **1** for explanatory convenience, and configurations of the floors **11** of the eighth floor and the ninth floor are configured similar to the floor **11** of the seventh floor.

(38) FIG. **2** is a top view schematically illustrating an example of a structure of the floor **11** of the rack **10** of the automated storage and retrieval system **1** according to the present embodiment. In the present embodiment, each floor **11** of the rack **10** defines a plurality of sections **13** which are aligned along the xy plane. The section **13** defines, for example, outlines of a square or a rectangle in plan view. In the present embodiment, the support columns **12** are respectively arranged at four corners of the one section **13**. The one storage bin **20** occupies the one section **13**. Similarly, the one transportation robot **30** occupies the one section **13**. That is, an outline of the storage bin **20** and an outline of the transportation robot **30** are respectively arranged in the outline of the one section **13** in plan view. Furthermore, an interval between a pair of the mutually neighboring support columns **12** and **12** is set larger than widths of the transportation robot **30** and the storage bin **20** defined in the x axis direction and a y axis direction.

(39) Each floor **11** defines storage areas **14** including the sections **13** in which the storage bins **20** are aligned, and movement passages **15** of the transportation robots **30** including the sections **13** other than the storage areas **14**. In the illustrated example, for example, two rows of bin groups of the plurality of storage bins **20** aligned along the y axis direction are aligned in the x axis direction in the storage area **14**. That is, all of the storage bins **20** making up the bin group of these two rows of the storage bins **20** face the movement passages **15** at all times. On the other hand, the transportation robots **30** can run on the movement passages **15**. As described later, the transportation robot **30** can also run in the storage area **14**, through a space below a bottom surface of the storage bin **20**.

(40) FIG. **3** is a perspective view schematically illustrating an example of a structure of the rack **10** of the automated storage and retrieval system **1** according to the embodiment. Referring to FIGS. **1** to **3** together, the rack **10** includes one or more transportation elevators **16**. The transportation elevators **16** can make reciprocating movement in the z axis direction from the first floor to the ninth floor of the floors **11**, and stop each floor **11**. Each transportation elevator **16** is arranged in one section in the movement passage **15** of the transportation robot **30** on each floor **11**. In the present embodiment, the plurality of transportation elevators **16** may be provided to the rack **10**. The transportation elevator **16** can have only the transportation robot **30** get thereon or have the transportation robot **30** holding the storage bin **20** get thereon to move to each floor of the first floor to the ninth floor.

(41) The transportation elevator **16** includes, for example, shafts of four corners attached to the rack **10** and extend in the z axis direction, four roller chains attached to the respective shafts, and extend in the z axis direction, four sprockets enmeshing with the respective chains, and two electric motors (both of which are not illustrated) for respectively rotating and driving, for example, the two sprockets. The sprockets enmesh with the roller chains in response to rotation of the sprockets caused by the power of electric motors to ascend and descend the transportation elevator **16**. Note that the shafts may be formed by the support columns **12** of the rack **10**. The above mechanism is a mere example, and an arbitrary another mechanism realizing vertical movement of the transportation elevator **16** may be used.

(42) FIG. **4** schematically illustrates an example of the structure of the storage bin **20** of the automated storage and retrieval system **1** according to the embodiment. In the present embodiment,

the storage bin **20** includes a bin main body **21** defining an internal space of, for example, a cuboid, and four support legs **22** extending downward from four corners of a bottom surface of the bin main body **21**. The bin main body **21** includes four sidewalls **21a** surrounding four sides, an opening part **21b** opened along upper ends of the sidewalls **21a**, and a bottom surface **21c** closed along lower ends of the sidewalls **21a**. Although the opening part **21b** is opened in the present embodiment, the opening part **21b** may be closed with, for example, a lid or a cover. The sidewalls **21a** extend along the xz plane or the yz plane. Furthermore, the bottom surface **21c** extends along the xy plane. The storage bin **20** is in contact with the floor **11** with the four support legs **22**. The storage bin **20** may be formed by, for example, a resin material. Furthermore, the storage bin **20** may be a foldable bin. A size of the storage bin **20** is preferably set appropriately based on a size of the rack **10** or each floor **11** or a size of an item **23**.

(43) The one or more items **23** are contained in the internal space in the bin main body **21**. The item **23** can be identified by, for example, a unique Stock Keeping Unit (SKU) set to the item **23**. Although the item **23** is, for example, one unit of a product or the like, the item **23** may be a so-called case product packaged in a unit of a plurality of identical products. Furthermore, the one storage bin **20** may contain only the items **23** of one type, or may contain the items **23** of a plurality of types. The storage bin **20** can be identified based on a unique ID set to each storage bin **20**. The unique ID of this storage bin **20** is managed in correspondence with the SKU of the item **23** contained in the storage bin **20**.

(44) FIG. 5 is a perspective view schematically illustrating an external appearance of the transportation robot **30** of the automated storage and retrieval system **1** according to the embodiment. In the present embodiment, a plurality of the transportation robots **30** is preferably arranged on each floor **11** of the rack **10**. The transportation robot **30** is, for example, an autonomous running transportation robot including a thin housing **31** of a substantially cuboid shape. The housing **31** is formed by, for example, a resin material. The transportation robot **30** can run along a line **17** (see, for example, FIG. 2) by tracing the line **17** drawn on the floor **11** of the rack **10** (line tracing function). The line **17** is lines of two directions which extend passing center positions of the x axis direction and the y axis direction in, for example, each section **13**, and are perpendicular to each other at a center of the section **13**. These lines **17** are drawn in all of the sections **13** including the storage areas **14**, the movement passages **15**, and the transportation elevators **16** of the floor **11**.

(45) An upper surface of the housing **31** of the transportation robot **30** extends flat along the xy plane. A height of the transportation robot **30** is set smaller than the height of the support leg **22** of the storage bin **20**. Furthermore, an outline of the housing **31** in top view is defined substantially as, for example, a square. Similarly, the length of one side of the transportation robot **30** is set smaller than the length of each side of the storage bin **20**. That is, the transportation robot **30** can enter the space below the bin main body **21** from between the pair of mutually neighboring support legs **22** of the storage bin **20**. In this regard, the size of the transportation robot **30** is preferably set appropriately based on sizes of the rack **10**, each floor **11**, and the storage bin **20**.

(46) The transportation robot **30** can change the height thereof between a first configuration where the height of the upper surface of the housing **31** is set to a first height, and a second configuration where the height is set to a second height higher than the first height. As described above, the transportation robot **30** of the first configuration can enter the space below the bin main body **21** from between the pair of mutually neighboring support legs **22** of the storage bin **20**. In this case, when the transportation robot **30** changes the height of the housing **31** from the first configuration to the second configuration, the transportation robot **30** holds the storage bin **20** on the upper surface of the housing **31** as illustrated in FIG. 6. As a result, the storage bin **20** can be lifted from the floor **11**. The transportation robot **30** can run in both of the first configuration and the second configuration. That is, the transportation robot **30** can run on the floor **11** while lifting the storage bin **20**.

(47) Back to FIG. 1, in the present embodiment, the picking stations **40** are established along, for example, the peripheral edge of the surface of the floor **11** of the rack **10**. That is, the picking station **40** is established in, for example, the rack **10**. In the present embodiment, for example, the two picking stations **40** are established on the floor **11** of the second floor of the rack **10**. The picking station **40** is a station for picking the item **23** from the storage bin **20** transported by the transportation robot **30** from the rack **10**. The operator **60** can execute picking work (delivery work) taking a standing posture on, for example, the surface of the floor **11** of the first floor. Note that details of a configuration of the picking station **40** will be described later.

(48) The management server **50** manages all of the rack **10**, the storage bins **20**, the transportation robots **30**, the transportation elevators **16**, and the picking stations **40** for warehousing, storage, and delivery of the automated storage and retrieval system **1**. This management is realized when a program stored in a storage unit is executed by a control unit as described later. More specifically, operations and processing of the transportation robots **30**, the transportation elevators **16**, and the picking stations **40** are executed according to information processing described in the program. That is, the information processing described in the program functions as specific means collaborated by software in correspondence with the program, and various hardware resources of the automated storage and retrieval system **1** when the program is read by the control unit.

(49) FIG. 7 is a functional block diagram schematically illustrating a configuration of the management server **50**. As illustrated in FIG. 7, the management server **50** includes a control unit **51** and a storage unit **52**. The control unit **51** includes a communication control unit **511**, a stock management unit **512**, and a device control unit **513**. On the other hand, the storage unit **52** stores a program **521**, rack information **522**, item information **523**, bin information **524**, and device information **525**. The control unit **51** manages the automated storage and retrieval system **1** by executing the program **521** stored in the storage unit **52**. This management server **50** may be realized on a physical server, yet may be realized on, for example, a cloud server.

(50) The communication control unit **511** controls communication between the management server **50** and at least the transportation robot **30** and a terminal of the operator **60** at the picking station **40**. A communication method may be, for example, Wi-Fi (registered trademark), Ethernet (registered trademark), optic or the like. The stock management unit **512** manages a stock status of the automated storage and retrieval system **1**. More specifically, the stock management unit **512** associates and manages information (SKU) for identifying each item **23**, information related to the number of stocks of each item **23** specified based on the SKU, information (ID) for identifying the storage bin **20** in which the item **23** is stored, and information related to a position of the section **13** of the floor **11** in which the storage bin **20** is stored. Each of these pieces of information is stored as the rack information **522**, the item information **523**, and the bin information **524** in the storage unit **52**.

(51) The device control unit **513** manages and controls at least states and operations of the transportation robots **30** and the picking stations **40**. More specifically, the device control unit **513** associates and manages information for identifying the transportation robot **30**, information related to a current state of the transportation robot **30**, i.e., a charging state of the transportation robot **30**, information related to whether or not the transportation robot **30** engages in transportation, information related to order processing in which the transportation robot **30** engages in a case where the transportation robot **30** engages in the transportation, and information related to a current position of the transportation robot **30** at the floor **11** of the rack **10** or the picking station **40**. The information related to the order processing includes, for example, information related to transportation of which item **23** of the order processing the transportation robot **30** engages in. Each of these pieces of information is stored as the device information **525** in the storage unit **52**. Furthermore, the device control unit **513** associates and manages information for identifying the picking station **40**, and information related to an order for which picking work is executed at the picking station **40**. Each of these pieces of information is stored as the device information **525** in

the storage unit **52**.

(52) Furthermore, the device control unit **513** generates commands for the transportation robot **30** and the transportation elevator **16** per order processed by the automated storage and retrieval system **1**. More specifically, the device control unit **513** specifies the storage bin **20** for containing the item **23** designated by the order based on each of the above pieces of information, and specifies the transportation robot **30** which needs to be assigned to transport the storage bin **20**. The device control unit **513** specifies a movement route **R1** (referred to as a “robot route” below) of the transportation robot **30** to the section **13** in which the storage bin **20** is stored, and a movement route **R2** (referred to as a “bin route” below) of the transportation robot **30** from the section **13** to the picking station **40** which is a destination. These pieces of generated information are transmitted as commands to the transportation robot **30** via the communication control unit **511**.

(53) FIG. **8** is a functional block diagram schematically illustrating a configuration of the transportation robot **30**. As illustrated in FIG. **8**, the transportation robot **30** includes a control unit **32** and a mechanism unit **33**. The control unit **32** includes a communication control unit **321**, a running control unit **322**, and a lifting control unit **323**. The mechanism unit **33** includes a communication unit **331**, a plurality of driving wheels **332**, a plurality of driving motors **333**, a lifting mechanism **334**, a sensor **335**, and a battery **336**. The control unit **32** controls an operation of the transportation robot **30** by executing a program (not illustrated) stored in the storage unit (not illustrated). The program may be stored in a physical storage unit incorporated in the transportation robot **30**, yet may be stored on, for example, a cloud server.

(54) The communication unit **331** establishes communication between at least the management server **50**, the transportation elevator **16**, and the terminal of the operator **60** at the picking station **40**. A communication method may be, for example, Wi-Fi (registered trademark), Ethernet (registered trademark), optic or the like. Communication performed by the communication unit **331** is controlled by the communication control unit **321**. The driving wheel **332** is a wheel for realizing running of the transportation robot **30**. The plurality of driving motors **333** drive the driving wheels **332** and the lifting mechanism **334**. Driving of the driving wheels **332** is controlled by the running control unit **322**. The lifting mechanism **334** establishes the above-described first configuration and second configuration by lifting the housing **31** of the transportation robot **30**. An operation of this lifting mechanism **334** is controlled by the lifting control unit **323**.

(55) The sensor **335** is, for example, an optical sensor for realizing the above-described line tracing function. More specifically, the sensor **335** is used to control running of the transportation robot **30** along the line **17** by reading a boundary of the line **17** drawn on each section **13**. Furthermore, although the two strips of the lines **17**, **17** cross at the center of each section **13**, the sensor unit **337** reads the line **17** perpendicular to the line **17** along which the transportation robot **30** is running, so that the transportation robot **30** can specify the center position of each section **13**. Thus, the transportation robot **30** can stop at, for example, the center position of each section **13**. The battery **336** is, for example, a rechargeable battery. On each floor **11** of the rack **10**, one or more charging spots (not illustrated) which enable charging of the battery **336** of the transportation robot **30** may be formed.

(56) FIG. **9** is a side view of the rack **10** schematically illustrating an example of the configuration of the picking station **40** of the automated storage and retrieval system **1** according to the embodiment. FIG. **10** is a plan view of the rack **10** schematically illustrating the example of the configuration of the picking station **40** of the automated storage and retrieval system **1** according to the embodiment. As illustrated in FIG. **9**, the picking station **40** is established continuing to this floor **11** at the peripheral edge of the floor **11** of the second floor of the rack **10**. More specifically, the picking station **40** is formed on the surface of the floor **11** of the second floor which extends in, for example, the y axis direction toward an outer side of an outer peripheral edge of the floors **11** of the third floor to the ninth floor. In the present embodiment, similar to the floor **11** of the second floor, the floor **11** of the first floor also extends toward the outer side of the outer peripheral edges

of the floors **11** of the third floor to the ninth floor. The operator **60** stands on the surface of the floor **11** of the first floor, and does work of picking the item **23** from the storage bin **20**. The height of the surface of the floor **11** of the second floor is suitable for the operator **60** whose height is, for example, 165 cm which is an average height of adults to do the work.

(57) As illustrated in FIG. **10**, the picking station **40** defines a plurality of picking positions **41A** to **41D** at which the storage bins **20** used for picking work of the item **23** are arranged on the floor **11** of the second floor. In the present embodiment, for example, the four picking positions **41A** to **41D** are aligned adjacent to each other in the x axis direction along the peripheral edge of the second floor **11** of the rack **10**. Each of the picking positions **41A** to **41D** occupies the one section **13** of the floor **11**. The picking positions **41A** to **41D** are aligned in front of the operator **60** facing the rack **10**, and the picking positions **41B** and **41C** of the picking positions **41A** to **41D** directly face the operator **60**. As is apparent from FIG. **9**, the upper floors **11** do not extend to a space above the picking positions **41A** to **41D** in the z axis direction. In this regard, the upper floors **11** may extend to the space above the picking positions **41A** to **41D** in the z axis direction.

(58) On the floor **11** of the second floor, for example, the two sections **13** extend toward the outer side in the y axis direction from the picking positions **41A** and **41D**. These four sections **13** in total define loading positions **43A** to **43D** at which the items **23** are loaded to shipment bins **42** which need to be shipped to clients of orders. Each of the loading positions **43A** to **43D** occupies the one section **13** of the floor **11**. The shipment bin **42** employs the same configuration as that of the storage bin **20**. According to the present embodiment, all of the loading positions **43A** to **43D** are aligned to a side of the operator **60** who faces the rack **10**, and directly face the operator **60**. Thus, as is clear from FIG. **10**, the four picking positions **41A** to **41D** and the four loading positions **43A** to **43D** are aligned surrounding the operator **60** on the floor **11** of the second floor.

(59) Furthermore, the sections **13** further surrounding the four picking positions **41A** to **41D** and the four loading positions **43A** to **43D** define the movement passage **15** of the transportation robot **30** on the floor **11** of the second floor. Although FIG. **10** omits illustration of the lines **17** for ease of description, the lines **17** are drawn in all of the sections **13** in which the four picking positions **41A** to **41D** and the four loading positions **43A** to **43D**, and the movement passages **15** are formed. Thus, as indicated by arrows in FIG. **10**, the transportation robot **30** can enter the four picking positions **41A** to **41D** and the four loading positions **43A** to **43D** from the movement passages **15**. Note that the storage bin **20** is not arranged in the sections **13** forming the movement passages **15**. Thus, the transportation robot **30** can transport the storage bin **20** between the storage area **14** of the rack **10** and the four picking positions **41A** to **41D**. On the other hand, the transportation robot **30** can transport the shipment bin **42** on which the item **23** has been loaded, to, for example, a shipping station (not illustrated) established in the rack **10**.

(60) Although the storage bins **20** are aligned on each floor **11** of the rack **10** as is apparent from FIG. **9**, the storage bins **20** for containing the items **23** of higher inventory turnover ratios are preferably aligned on the lower floors **11**. In other words, the storage bins **20** for containing the items **23** of lower inventory turnover ratios are preferably aligned in the upper floors **11**. In the present embodiment, the picking stations **40** are established on the floor **11** of the second floor, and therefore the items **23** of lower inventory turnover ratios are stored on the upper floors **11** for which physical movement distances of the transportation robots **30** from the second floor are assumed to be long. Similarly, the items **23** of higher inventory turnover ratios are stored on the floors **11** which are close to the second floor and to which physical movement distances are assumed to be short, or the second floor for which the transportation elevator **16** does not need to be used. Furthermore, the picking stations **40** are not limited to the picking stations provided adjacent to the floor **11** of the second floor. The picking stations **40** may be provided adjacent to the other floors **11** likewise according to installation environment such as a shape, a size, and the like of a warehouse in which the automated storage and retrieval system **1** is installed, or a control mode or the like of the automated storage and retrieval system **1**. Furthermore, the automated storage and retrieval

system **1** may be provided with the plurality of picking stations **40**. Furthermore, the picking positions are not limited to the above-described four picking positions **41A** to **41D**, and the loading positions are not limited to the above-described four loading positions **43A** to **43D**.

(61) Next, a configuration of the rack **10** will be specifically described. FIG. **11** is a perspective view illustrating a schematic configuration of a rack module **2** included in the rack **10**. The rack **10** includes the plurality of rack modules **2**, and is formed by connecting the plurality of rack modules **2**. Furthermore, as described above, the rack **10** stores the plurality of storage bins **20**, and includes the plurality of floors **11** allowing the transportation robots **30** to run along surfaces thereof. The rack module **2** includes a floor module **80** which is part of each floor **11**, and a plurality of support column modules **70** supporting the floor module **80**. A configuration of the rack module **2** will be specifically described below.

(62) The rack module **2** is a minimum unit structure making up the rack **10**, and the rack **10** is formed by the plurality of rack modules **2**. More specifically, the rack **10** is formed by coupling the plurality of rack modules **2** in a horizontal direction and the vertical direction. The rack module **2** forms a zone space **S** in the rack **10**. The zone space **S** is a space corresponding to the section **13** of the floor **11**, and is a space between the neighboring sections **13** in the **z** direction (see FIG. **3**). That is, each zone space **S** is a space in which the one storage bin **20** is stored, and is a minimum unit space making up the storage area **14**, and each zone space **S** is a minimum unit space making up the movement passage **15**. Note that, although part of components is shared between the neighboring rack modules **2**, the shared components are also components of each rack module **2**.

(63) More specifically, as illustrated in FIG. **11**, each rack module **2** includes part of the eight respective support column modules **70** and the floor module **80**. FIGS. **12A** and **12B** are perspective views of the support column module **70**, and FIGS. **13A** to **13F** are respectively a plan view, a right side view, a left side view, a top view, a bottom view, and a back view of the support column module **70**. Furthermore, FIG. **14** is a perspective view of the floor module **80**. Furthermore, FIG. **13G** is an enlarged view of FIG. **13A**.

(64) As illustrated in FIGS. **12A**, **12B**, and **13A** to **13F**, the support column module **70** is a member having a columnar shape extending in an axial line **a1** direction and having rigidity, and includes a main body **71**, an insertion plug **72**, and an accommodation plug **73**. The main body **71** includes, for example, a base part **71a** which is a bar-shaped part extending with the axis **a1** arranged as the center axis or the substantial center axis, a plurality of rib parts **71b** which is plate-like parts extending in the axis **a1** direction, and a plurality of rib parts **71c** which is plate-like parts extending in a direction perpendicular to the axis **a1**. The base part **71a**, the plurality of rib parts **71b**, and the plurality of rib parts **71c** are integrally connected to form the support column module **70**.

(65) As illustrated in FIGS. **12A** and **12B**, the insertion plug **72** is connected to one end in the axis **a1** direction of the main body **71** at which an insertion part **74** protruding toward one side of the axis **a1** direction is formed. The insertion part **74**, for example, includes four insertion protrusion parts **74a** which are parts protruding in the axis **a1** direction. The insertion protrusion parts **74a** are arranged at equal intervals with the axis **a1** arranged at the center, and for example, the two insertion protrusion parts **74a** are respectively aligned in a direction perpendicular to the axis **a1** to form two rows of the insertion protrusion parts **74a**. These two rows face each other. Note that the number of the insertion protrusion parts **74a** provided at the insertion plug **72** is not limited to four, and may be one or other plural numbers. The number of the insertion protrusion parts **74a** provided at the insertion plug **72** is determined to match with, for example, the number of accommodation recess parts **75a** of the accommodation plug **73** described below.

(66) As illustrated in FIG. **13F**, each insertion protrusion part **74a** has, for example, a cylindrical shape extending along the axis **a1**, and, more specifically, has, for example, a square cylindrical shape whose corner edge is a curved surface having an arc-shaped cross section. Furthermore, claw parts **74c** extending along the axis **a1** are formed at a distal end **74b** of each insertion protrusion

part **74a**. The claw part **74c** has a shape which can engage with a corresponding part, and has, for example, a shape which can engage with a corresponding part by snap-fit. For example, the two claw parts **74c** are provided at each insertion protrusion part **74a**.

(67) As illustrated in FIGS. **12A** and **12B**, the accommodation plug **73** is connected to an other end in the axis **a1** direction of the main body **71**, and the insertion plug **72** and the accommodation plug **73** of the support column module **70** oppose to each other in the axis **a1** direction. The accommodation plug **73** includes an accommodation part **75**, and the accommodation part **75** can accommodate the insertion part **74** of the insertion plug **72** of the support column module **70**. More specifically, the accommodation part **75** includes the four accommodation recess parts **75a** which can respectively accommodate the four insertion protrusion parts **74a** of the insertion part **74** of the insertion plug **72**. As illustrated in FIG. **13A**, the accommodation recess part **75a** has a shape matching the insertion protrusion part **74a**, and is a recess part recessed toward the insertion plug **72** side in the axis **a1** direction. Each accommodation recess part **75a** can accommodate the corresponding insertion protrusion part **74a**. Each accommodation recess part **75a** may be able to accommodate the corresponding insertion protrusion part **74a** by clearance fit, accommodate the corresponding insertion protrusion part **74a** by stationary fit, or accommodate the corresponding insertion protrusion part **74a** by transition fit. Note that the number of the accommodation recess parts **75a** provided to the accommodation plug **73** is not limited to four, and may be one or other plural numbers. The number of the insertion protrusion parts **74a** provided to the insertion plug **72** may be determined according to, for example, the number of the floor modules **80** supported by the support column module **70**.

(68) More specifically, the four accommodation recess parts **75a** are arranged to meet the four insertion protrusion parts **74a** of the insertion part **74**, and are arranged at equal intervals with the axis **a1** arranged at the center, and the two accommodation recess parts **75a** are respectively aligned in a direction perpendicular to the axis **a1** to form two rows of the accommodation recess parts **75a** facing each other. Furthermore, each accommodation recess part **75** is a recess part defining a bottomed cylindrical surface extending along the axis **a1**, has a square cross-sectional outline perpendicular to the axis **a1** as illustrated in, for example, FIGS. **12A**, **13A**, and **13G**, and can accommodate the corresponding insertion protrusion part **74a** by stationary fit. Furthermore, at a bottom part **75b** of each accommodation recess part **75a**, engagement recess parts **75c** extending along the axis **a1** are formed. Each engagement recess part **75c** can accommodate the claw part **74c** of the corresponding insertion protrusion part **74a**, and can engage with the accommodated claw part **74c**. Each engagement recess part **75c** has, for example, a shape which can engage with the corresponding claw part **74c** by snap-fit. Each accommodation recess part **75a** is provided with, for example, the two engagement recess parts **75c**.

(69) Furthermore, as illustrated in, for example, FIGS. **12A**, **13A**, and **13G**, the accommodation plug **73** is formed with a floor reception part **76** formed around the accommodation part **75**. The floor reception part **76** is a part to support the floor module **80** in the axis **a1** direction. The floor reception part **76** includes a flange part **76a** forming a face (floor reception face **76b**) extending along a plane perpendicular to the axis **a1**, and engagement protrusion parts **76c** protruding from the flange part **76a** to a side opposite to the main body **71** in the axis **a1** direction. As illustrated in, for example, FIG. **13G**, the floor reception face **76b** extends in parallel to the plane perpendicular to the axis **a1**. The engagement protrusion part **76c** has a shape which can engage with a corresponding part, and includes, for example, a claw part **76d** having a shape which can engage with a corresponding part by snap-fit, and pins **76e** which can be accommodated in corresponding parts. The floor reception part **76** is provided with, for example, the plurality of claw parts **76d**, and is provided with the plurality of pins **76e**.

(70) The accommodation part **75** protrudes from the flange part **76a** of the floor reception part **76** toward a side opposite to the main body **71** in the axis **a1** direction. A height **h1** (see FIGS. **13B** to **13E**) of the accommodation part **75** is a thickness **t1** (see FIG. **14**) of the floor module **80** or more.

Note that the height **h1** of the accommodation part **75** is a distance in the axis **a1** direction between the floor reception face **76b** of the flange part **76a** and a distal end face **75d** of the accommodation part **75**. Note that the height **h1** of the accommodation part **75** may be smaller than the thickness **t1** of the floor module **80**. As illustrated in FIGS. **12A** and **13G**, the accommodation part **75** includes plate-like inner wall parts **75e**, and an outer wall part **75f** surrounding these four inner wall parts **75e**. These inner wall parts **75e** and outer wall part **75f** define the four accommodation recess parts **75a**. The inner wall part **75e** is, for example, a plate-like part extending in the direction perpendicular to the axis **a1**, and the outer wall part **75f** is, for example, an annular cylindrical part around the axis **a1**. An outer circumferential surface **75g** of the outer wall part **75f** is a cylindrical surface, and has, for example, a square or substantial square outline in plan view as illustrated in FIG. **13G**. Note that the shape of the outer circumferential surface **75g** of the outer wall part **75f** is not limited to a shape having the square outline in plan view, and may have a shape having an outline of another shape in plan view. For example, the shape of the outer circumferential surface **75g** of the outer wall part **75f** may be other polygons or circles in plan view, or may be a shape substantially having an outline of these shapes.

(71) The main body **71**, the insertion plug **72**, and the accommodation plug **73** are parts of the support column module **70** which are integrally formed by the same material, and the main body **71**, the insertion plug **72**, and the accommodation plug **73** are integrally formed. A material of the support column module **70** is, for example, a resin. The material of the support column module **70** is not limited to the resin, and may be, for example, metal.

(72) The support column module **70** employs the above-described configuration, and the two support column modules **70** are coupled to each other by accommodating and connecting the insertion plug **72** of the one support column module **70** in and to the accommodation plug **73** of the other support column module **70**. Furthermore, at a time of this coupling, the claw parts **74c** of each insertion protrusion part **74a** of the insertion plug **72** of the one support column module **70** are accommodated in the corresponding engagement recess parts **75c** of the corresponding accommodation recess part **75a** of the accommodation plug **73** of the other support column module **70**, and these claw parts **74c** and engagement recess parts **75c** are engaged by snap-fit. On the other hand, the two support column modules **70** coupled as described above are detached by pulling the insertion plug **72** of the one support column module **70** from the accommodation plug **73** of the other support column module **70**.

(73) The plurality of support column modules **70** are coupled in this way, so that the one support column **12** can be formed. When the support column modules **70** are coupled, each insertion protrusion part **74a** fits to the corresponding accommodation recess part **75a**, and the claw parts **74c** and the engagement recess parts **75c** engage with each other. Consequently, it is possible to increase coupling strength between these support column modules **70**, and increase strength of the support column **12**. On the other hand, as described above, it is possible to decouple the support column modules **70** of the support column **12**.

(74) The floor module **80** is a plate-like member corresponding to the one section **13** of the floor **11**, and includes a surface **81** which is a face on which the section **13** is formed. The floor module **80** includes a plurality of corner portions. As described above, the section **13** defines, for example, the square or rectangular outline in plan view, and, similarly, as illustrated in FIG. **14**, the floor module **80** defines, for example, a square or rectangular outline in plan view, and includes four corner portions **83a** to **83d**. The thickness **t1** of the floor module **80** is a distance between the surface **81** and a back surface **82** which is a face opposing to the surface **81**. The corner parts of the floor module **80** can be supported by the accommodation plugs **73** of the support column modules **70**. For example, the floor module **80** has a structure in which the floor modules **80** the number of which is the same as the number of the corner portions included in the floor module **80** can be supported by the accommodation part **75** and the floor reception part **76** of the one accommodation plug **73**. Note that the floor module **80** may have a structure in which the floor modules **80** the

number of which is different from the number of corner portions included in the floor module **80** can be supported by the accommodation part **75** and the floor reception part **76** of the one accommodation plug **73**.

(75) Respectively at the four corner portions **83a** to **83d** of the floor module **80**, cutout faces **84a** to **84d** are formed. Note that the corner portions **83a** to **83d** are aligned clockwise in order in plan view. The cutout faces **84a** to **84d** are each a part defining, at each of the corner portions **83a** to **83d**, a space recessed in an extension direction of the floor module **80**, and is a face facing this space. The cutout faces **84a** to **84d** are each a face corresponding to part of the outer circumferential surface **75g** of the outer wall part **75f** of the accommodation part **75** of the support column module **70**. For example, the cutout faces **84a** to **84d** are each a face which can come into contact with part of the outer circumferential surface **75g** of the outer wall part **75f** of the accommodation part **75**, or a face facing a fixed or substantially fixed interval apart from each other.

(76) More specifically, for example, the cutout face **84a** is a face corresponding to a $\frac{1}{4}$ or substantially $\frac{1}{4}$ part of the outer circumferential surface **75g** of the outer wall part **75f** around the axis **a1**. The cutout face **84b** is a face corresponding to another $\frac{1}{4}$ or substantially $\frac{1}{4}$ part of the outer circumferential surface **75g** of the outer wall part **75f** around the axis **a1**. The cutout face **84c** is a face corresponding to another $\frac{1}{4}$ or substantially $\frac{1}{4}$ part of the outer circumferential surface **75g** of the outer wall part **75f** around the axis **a1**. The cutout face **84d** is a face corresponding to another $\frac{1}{4}$ or substantially $\frac{1}{4}$ part of the outer circumferential surface **75g** of the outer wall part **75f** around the axis **a1**. Note that the $\frac{1}{4}$ part of the outer circumferential surface **75g** of the outer wall part **75f** which each of the cutout faces **84a** to **84d** faces is aligned clockwise in order around the axis **a1**. Hence, when the four floor modules **80** are annularly aligned facing the same direction such that the surfaces **81** are coplanar, the cutout faces **84a** to **84d** of the four respective floor modules **80** form through-holes surrounding outer circumferential surfaces **75g** of the accommodation parts **75**. That is, the cutout face **84a** of the one floor module **80**, the cutout face **84b** of the one floor module **80**, the cutout face **84c** of the one floor module **80**, and the cutout face **84d** of the one floor module **80** are annularly aligned in this order. Note that one cutout face may not be the face corresponding to the $\frac{1}{4}$ or substantially $\frac{1}{4}$ part of the outer circumferential surface **75g** of the outer wall part **75f** around the axis **a1**, and may be a face corresponding to a part of another size of the outer circumferential surface **75g**. According to, for example, the number of the corner portions of the floor module **80**, the number of the floor modules **80** supported by the one accommodation plug **73**, or the like, the one cutout face may be a face corresponding to a part of a predetermined size of the outer circumferential surface **75g**.

(77) The $\frac{1}{4}$ or substantially $\frac{1}{4}$ part of the outer circumferential surface **75g** of the outer wall part **75f** around the axis **a1** which the cutout faces **84a** to **84d** face is a part of the outer circumferential surface **75g** of the outer wall part **75f** facing the one accommodation recess part **75a**. More specifically, as illustrated in FIG. 13G, the part of the outer circumferential surface **75g** which the cutout face **84a** faces is a part of the outer circumferential surface **75g** facing the one accommodation recess part **75a** along a dotted line **p1**. Furthermore, the part of the outer circumferential surface **75g** which the cutout face **84b** faces is a part of the outer circumferential surface **75g** facing the one accommodation recess part **75a** along a dotted line **p2**. Furthermore, the part of the outer circumferential surface **75g** which the cutout face **84c** faces is a part of the outer circumferential surface **75g** facing the one accommodation recess part **75a** along a dotted line **p3**. Furthermore, the part of the outer circumferential surface **75g** which the cutout face **84d** faces is a part of the outer circumferential surface **75g** facing the one accommodation recess part **75a** along a dotted line **p4**.

(78) Furthermore, as illustrated in FIG. 15, near the four cutout faces **84a** to **84d** of the floor module **80**, an engagement recess part **85** and recess parts **86** which can respectively accommodate the claw part **76d** and the pins **76e** of the engagement protrusion part **76c** provided to the floor

reception part **76** of the support column module **70** are formed. The engagement recess part **85** can engage with the accommodated claw part **76d** by snap-fit. The pin **76e** is accommodated in the recess part **86** to position the floor module **80** with respect to the support column module **70**. In a state where the cutout faces **84a**, **84b**, **84c**, or **84d** faces the corresponding part **p1**, **p2**, **p3**, or **p4** of the outer circumferential surface **75g** of the outer wall part **75f** of the accommodation part **75** in desired forms, the claw part **76d** and the pins **76e** may be respectively accommodated in the corresponding engagement recess part **85** and the recess parts **86**. Note that, although the engagement recess part **85** and the recess parts **86** continue in the illustrated example, the forms of the engagement recess part **85** and the recess parts **86** are not limited to these.

(79) Furthermore, as illustrated in FIG. **14**, the floor module **80** includes a coupling part **87** for coupling to the other floor modules **80**. The coupling part **87** includes, for example, the engagement protrusion parts **87a** provided on one side surfaces of respective surfaces (side surfaces **80a** and **80c** and side surfaces **80b** and **80d**) of a pair of two opposing side surfaces of the floor module **80**, and the engagement recess parts **87b** provided on other side surfaces of respective surfaces (the side surfaces **80a** and **80c** and the side surfaces **80b** and **80d**) of a pair of two opposing side surfaces. The engagement protrusion part **87a** is a protrusion protruding from the side surface, and the engagement recess part **87b** is a recess part recessed from the side surface. When the floor module **80** is aligned at a desired position adjacent to the other floor module **80**, the engagement protrusion part **87a** is accommodated in the corresponding engagement recess part **87b**. In the illustrated example, the side surface **80a** and the side surface **80b** of the floor module **80** are provided with the engagement protrusion parts **87a**, and the side surface **80c** and the side surface **80d** of the floor module **80** are provided with the engagement recess parts **87b**. Furthermore, although the two engagement protrusion parts **87a** or the two engagement recess parts **87b** are provided on one side surface, the numbers of the engagement protrusion parts **87a** and the engagement recess parts **87b** are not limited to these.

(80) Furthermore, on the surface **81** of the floor module **80**, the two lines **17** intersecting at a center or a substantial center of the surface **81** is provided. For example, the two lines **17** are perpendicular or substantially perpendicular to each other. As described above, the transportation robot **30** can run along the lines **17** or stop on the line **17** by tracing these lines **17**.

(81) Note that, although the floor module **80** is the plate-like member, the floor module **80** may be solid or have a cavity inside. For example, the floor module **80** may be formed by stretching a plate-like surface member on a surface of a framework assembled in a grid pattern. Furthermore, the number of the corner portions **83** of the floor module **80** is not limited to four, and the shape of the floor module **80** in plan view is not limited to the square or the rectangle, and may be other shapes such as other polygons and circles.

(82) As illustrated in FIGS. **11**, **16**, and **17**, the rack module **2** includes the respective accommodation plugs **73** of the four support column modules **70** meeting the four corner portions **83a** to **83d** of the floor module **80**, the respective main bodies **71** and insertion plugs **72** of the other four support column modules **70** respectively coupled to the accommodation plugs **73** of these four support column modules **70**, and the one floor module **80**. Note that FIGS. **16** and **17** are exploded perspective views of the rack module **2**, FIG. **16** illustrates that the four support column modules **70** and the other four support column modules **70** are detached, and FIG. **17** further illustrates that the respective accommodation plugs **73** of the four support column modules **70** are detached from the floor module **80**. In the rack module **2**, the respective accommodation plugs **73** of the four support column modules **70** are perpendicular or substantially perpendicular to the surface **81** of the floor module **80**, and respectively support the floor module **80** at the four corner portions **83a** to **83d**.

(83) More specifically, as illustrated in FIGS. **11**, **16**, and **17**, in the rack module **2**, the accommodation plug **73** of the one support column module **70A1** among the four support column modules **70** supports the floor module **80** at the corner portion **83a** of the floor module **80**. More

specifically, the corner portion **83a** of the floor module **80** comes into contact with part of the floor reception face **76b** (see FIG. **13G**) of the floor reception part **76** of the accommodation plug **73**, and the floor module **80** is supported by the support column module **70A1** at the corner portion **83a**. Part of the floor reception face **76b** coming into contact with the corner portion **83a** of the floor module **80** is a part of the floor reception face **76b** (along the dotted line p1 in FIG. **13G**) facing the one accommodation recess part **75a** of the accommodation plug **73**. Furthermore, the part (see FIG. **13G**) along the dotted line p1 of the outer circumferential surface **75g** of the outer wall part **75f** faces the cutout face **84a** of the corner portion **83a** of the floor module **80**. Furthermore, the pin **76e** near the part along the dotted line p1 of the outer circumferential surface **75g** of the accommodation recess part **75a** is accommodated in the recess part **86** of the corner portion **83a**, and the support column module **70A1** is positioned with respect to the floor module **80**. Furthermore, the claw part **76d** near the part along the dotted line p1 of the outer circumferential surface **75g** is accommodated in and engaged with the engagement recess part **85** of the corner portion **83a** by snap-fit, and the support column module **70A1** is fixed to the floor module **80**. Furthermore, the insertion plug **72** of one support column module **70A2** among the four other support column modules **70** is inserted in the accommodation plug **73** of the support column module **70A1**, and the support column module **70A2** is coupled to the support column module **70A1**. In this state, the claw parts **74c** of the insertion protrusion part **74a** of the insertion plug **72** of the support column module **70A2** are engaged with the engagement recess parts **75c** of the accommodation recess part **75a** of the accommodation plug **73** of the support column module **70A1** by snap-fit.

(84) Similarly, as illustrated in FIGS. **11**, **16**, and **17**, in the rack module **2**, the accommodation plug **73** of the one support column module **70B1** among the four support column modules **70** supports the floor module **80** at the corner portion **83b** of the floor module **80**. More specifically, the corner portion **83b** of the floor module **80** comes into contact with part of the floor reception face **76b** (see FIG. **13G**) of the floor reception part **76** of the accommodation plug **73**, and the floor module **80** is supported by the support column module **70B1** at the corner portion **83b**. Part of the floor reception face **76b** coming into contact with the corner portion **83b** of the floor module **80** is a part of the floor reception face **76b** (along the dotted line p2 in FIG. **13G**) facing the one accommodation recess part **75a** of the accommodation plug **73**.

(85) Furthermore, the part (see FIG. **13G**) along the dotted line p2 of the outer circumferential surface **75g** of the outer wall part **75f** faces the cutout face **84b** of the corner portion **83b** of the floor module **80**. Furthermore, the pin **76e** near the part along the dotted line p2 of the outer circumferential surface **75g** of the accommodation recess part **75a** is accommodated in the recess part **86** of the corner portion **83b**, and the support column module **70B1** is positioned with respect to the floor module **80**. Furthermore, the claw part **76d** near the part along the dotted line p2 of the outer circumferential surface **75g** is accommodated in and engaged with the engagement recess part **85** of the corner portion **83b** by snap-fit, and the support column module **70B1** is fixed to the floor module **80**. Furthermore, the insertion plug **72** of the one support column module **70B2** among the four other support column modules **70** is inserted in the accommodation plug **73** of the support column module **70B1**, and the support column module **70B2** is coupled to the support column module **70B1**. In this state, the claw parts **74c** of the insertion protrusion part **74a** of the insertion plug **72** of the support column module **70B2** are engaged with the engagement recess parts **75c** of the accommodation recess part **75a** of the accommodation plug **73** of the support column module **70B1** by snap-fit.

(86) Similarly, as illustrated in FIGS. **11**, **16**, and **17**, in the rack module **2**, the accommodation plug **73** of the one support column module **70C1** among the four support column modules **70** supports the floor module **80** at the corner portion **83c** of the floor module **80**. More specifically, the corner portion **83c** of the floor module **80** comes into contact with part of the floor reception face **76b** (see FIG. **13G**) of the floor reception part **76** of the accommodation plug **73**, and the floor module **80** is

supported by the support column module **70C1** at the corner portion **83c**. Part of the floor reception face **76b** coming into contact with the corner portion **83c** of the floor module **80** is a part of the floor reception face **76b** (along the dotted line **p3** in FIG. **13G**) facing the one accommodation recess part **75a** of the accommodation plug **73**. Furthermore, the part (see FIG. **13G**) along the dotted line **p3** of the outer circumferential surface **75g** of the outer wall part **75f** faces the cutout face **84c** of the corner portion **83c** of the floor module **80**. Furthermore, the pin **76e** near the part along the dotted line **p3** of the outer circumferential surface **75g** of the accommodation recess part **75a** is accommodated in the recess part **86** of the corner portion **83b**, and the support column module **70C1** is positioned with respect to the floor module **80**. Furthermore, the claw part **76d** near the part along the dotted line **p3** of the outer circumferential surface **75g** is accommodated in and engaged with the engagement recess part **85** of the corner portion **83c** by snap-fit, and the support column module **70C1** is fixed to the floor module **80**. Furthermore, the insertion plug **72** of the one support column module **70C2** among the four other support column modules **70** is inserted in the accommodation plug **73** of the support column module **70C1**, and the support column module **70C2** is coupled to the support column module **70C1**. In this state, the claw parts **74c** of the insertion protrusion part **74a** of the insertion plug **72** of the support column module **70C2** are engaged with the engagement recess parts **75c** of the accommodation recess part **75a** of the accommodation plug **73** of the support column module **70C1** by snap-fit.

(87) Similarly, as illustrated in FIGS. **11**, **16**, and **17**, in the rack module **2**, the accommodation plug **73** of the one support column module **70D1** among the four support column modules **70** supports the floor module **80** at the corner portion **83d** of the floor module **80**. More specifically, the corner portion **83d** of the floor module **80** comes into contact with part of the floor reception face **76b** (see FIG. **13G**) of the floor reception part **76** of the accommodation plug **73**, and the floor module **80** is supported by the support column module **70D1** at the corner portion **83d**. Part of the floor reception face **76b** coming into contact with the corner portion **83d** of the floor module **80** is a part of the floor reception face **76b** (along the dotted line **p4** in FIG. **13G**) facing the one accommodation recess part **75a** of the accommodation plug **73**. Furthermore, the part (see FIG. **13G**) along the dotted line **p4** of the outer circumferential surface **75g** of the outer wall part **75f** faces the cutout face **84d** of the corner portion **83d** of the floor module **80**. Furthermore, the pin **76e** near the part along the dotted line **p4** of the outer circumferential surface **75g** of the accommodation recess part **75a** is accommodated in the recess part **86** of the corner portion **83d**, and the support column module **70D1** is positioned with respect to the floor module **80**. Furthermore, the claw part **76d** near the part along the dotted line **p4** of the outer circumferential surface **75g** is accommodated in and engaged with the engagement recess part **85** of the corner portion **83d** by snap-fit, and the support column module **70D1** is fixed to the floor module **80**. Furthermore, the insertion plug **72** of the one support column module **70D2** among the four other support column modules **70** is inserted in the accommodation plug **73** of the support column module **70D1**, and the support column module **70D2** is coupled to the support column module **70D1**. In this state, the claw parts **74c** of the insertion protrusion part **74a** of the insertion plug **72** of the support column module **70D2** are engaged with the engagement recess parts **75c** of the accommodation recess part **75a** of the accommodation plug **73** of the support column module **70D1** by snap-fit.

(88) As described above, the rack module **2** includes the floor module **80**, the accommodation plugs **73** of the respective support column modules **70A1**, **70B1**, **70C1**, and **70D1** supporting the floor module **80**, and the respective main bodies **71** and insertion plugs **72** of the support column modules **70A2**, **70B2**, **70C2**, and **70D2** protruding from the surface **81** of the floor module **80**.

(89) Note that the respective main bodies **71** and insertion plugs **72** of the support column modules **70A2**, **70B2**, **70C2**, and **70D2** of the one rack module **2** are components of the rack module **2** adjacent to this rack module **2** in the rack **10**. Thus, the respective main bodies **71** and insertion plugs **72** of the support column modules **70A2**, **70B2**, **70C2**, and **70D2** of the one rack module **2** are components shared with the other rack module **2** (see FIGS. **19A** and **19B**).

(90) Furthermore, for the rack module **2** making up the lowermost floor **11** in the rack **10**, support column end modules **77** illustrated in FIGS. **18A** and **18B** are used as members supporting the floor module **80** instead of the support column modules **70** (support column modules **70A1**, **70B1**, **70C1**, and **70D1**) (see FIGS. **19A** and **19B**). The support column end module **77** is the same member as part of the accommodation plug **73** side of the support column module **70**, and includes the accommodation plug **73** and the main body **78** which is part of the accommodation plug **73** side of the main body **71**. An end face **78a** of the main body **78** is a face which is perpendicular or substantially perpendicular to the axis **a1**.

(91) Furthermore, the other floor module **80** is attached to the rack module **2** making up the uppermost floor **11** or the rack module **2** which is the uppermost part in the rack **10** (see FIGS. **19A** and **19B**). That is, similar to the floor module **80** of the rack module **2**, the other floor module **80** is attached to the flange parts **76a** of the respective accommodation plugs **73** of the support column modules **70** (support column modules **70A2**, **70B2**, **70C2**, and **70D2**) protruding from the surface **81** of the floor module **80**.

(92) A plurality of the above-described rack modules **2** is used and assembled arbitrarily to form the rack **10**. FIGS. **19A** and **19B** are perspective views illustrating an example of the plurality of rack modules **2** connected in the horizontal direction and the vertical direction. As illustrated in FIGS. **19A** and **19B**, the rack modules **2** (a rack module **2A** and a rack module **2B**, or the rack module **2A** and a rack module **2D**) adjacent in the horizontal direction (an X direction or a Y direction) share the respective accommodation plugs **73** of the two support column modules **70** supporting the floor module **80**, and share the respective main bodies **71** and insertion plugs **72** of the two support column modules **70** protruding from the floor module **80**. On the other hand, as illustrated in FIGS. **19A** and **19B**, the rack modules **2** (the rack module **2A** and a rack module **2C**) adjacent in the horizontal direction (a direction oblique to the X direction or the Y direction) share the accommodation plug **73** of the one support column module **70** supporting the floor module **80**, and share the main body **71** and the insertion plug **72** of the one support column module **70** protruding from the floor module **80**.

(93) As described above, the one rack module **2** forms the zone space **S** corresponding to the one storage bin **20**, so that the rack module **2** can make a component unit of the rack **10** smaller. Consequently, it is possible to adopt various forms of the rack **10**, and flexibly design the form of the automated storage and retrieval system **1** at various places at which the automated storage and retrieval system **1** is installed. Consequently, it is possible to form the rack **10** in a shape and a size matching a place at which the automated storage and retrieval system **1** is installed, and improve space efficiency of the rack **10**. For example, it is possible to connect part of the rack **10** to a space in a basement of a warehouse. Furthermore, it is possible to form the shape of the rack **10** in a complex shape, more efficiently form the movement passages **15** of the transportation robots **30** and the storage areas **14**, and improve transportation efficiency and storage efficiency of the item **23**. Furthermore, it is possible to more flexibly determine installation positions, shapes, and the like of the transportation elevator **16**. More specifically, it is possible to provide the transportation elevators **16** at arbitrary places of the rack **10** including the plurality of floors. Consequently, it is possible to change the lengths, the positions, and the like of the transportation elevators **16** according to, for example, a form of the rack **10**, various processing procedures of the automated storage and retrieval system **1**, and item liquidity at the automated storage and retrieval system **1**. Furthermore, unlike conventional automated storage and retrieval systems, it is possible to install the transportation elevators **16** at an inside of the rack **10**, too, which is not surroundings of the rack **10**.

(94) Furthermore, each rack module **2** includes the two crossing lines **17** which are minimum component units of the movement passage of the transportation robot **30**. Consequently, it is possible to form the movement passages of arbitrary shapes of the transportation robots **30** by arbitrarily connecting the plurality of rack modules **2**. Thus, the rack module **2** makes it possible to

form the movement passages of the transportation robots **30** in arbitrary shapes, and make the movement passages of the transportation robots **30** complicated.

(95) Furthermore, the rack module **2** can also make up the picking station **40**. Consequently, it is possible to more flexibly determine installation positions, shapes, sizes, and the like of the picking stations **40** in the rack **10**. The rack module **2** makes it possible to form the movement routes of the transportation robots **30** on each floor **11**, and form complicated movement routes. Consequently, it is possible to provide the picking stations **40** on the middle floors **11** in the rack **10**. In this case, for example, a robot arm can perform work of picking the item **23** at the picking station **40**.

Furthermore, for example, it is possible to provide the picking stations **40** near warehouse windows in a small warehouse, and perform work of picking the items **23** through the windows.

(96) Furthermore, it is possible to partially form the rack **10** at a place at which the automated storage and retrieval system **1** is installed, and improve installation efficiency of the rack **10**. For example, it is possible to disassemble an existing rack, and partially form the rack **10** in a space left as a result of disassembly. Furthermore, it is possible to easily increase the zone spaces **S** in the existing rack **10**, and easily disassemble the rack **10** and easily change a layout. Consequently, when the automated storage and retrieval system **1** causes a failure, it is possible to partially disassemble the rack **10** by detaching the rack module **2**, and easily and efficiently repair the automated storage and retrieval system **1**. Furthermore, it is possible to easily and efficiently maintain the automated storage and retrieval system **1**.

(97) Thus, the automated storage and retrieval system **1** according to the embodiment of the present disclosure enables flexible design of the form thereof.

(98) Note that the support column module **70** may not include the floor reception part **76**. FIG. **20** is a perspective view illustrating a support column module **70V** according to a modified example of the support column module **70**. As illustrated in FIG. **20**, the support column module **70V** may include hooks **79** fixed to the accommodation part **75** instead of the flange part **76a**. FIG. **21** is a perspective view illustrating an example of the hook **79**. The hook **79** includes the same functions as those of the floor reception face **76b** and the engagement protrusion part **76c** of the above-described floor reception part **76**. More specifically, for example, the hook **79** includes a floor engagement part **79a** as illustrated in FIGS. **20** and **21**. The floor engagement part **79a** can engage with the floor module **80** from below at each of the corner portions **83a** to **83d** of the floor module **80**, and support the floor module **80**. Furthermore, the hook **79** includes a support column engagement part **79b** as illustrated in FIGS. **20** and **21**. The support column engagement part **79b** engages with the accommodation part **75** of the support column module **70V** so as to meet each accommodation recess part **75a**, and is fixed thereto.

(99) As illustrated in, for example, FIGS. **20** and **21**, the floor engagement part **79a** of the hook **79** is a part bent from one end of a base part **79c** of the hook **79** to an other end side (upper side) and extending upward, and is formed such that the floor engagement part **79a** can engage with and be fixed to the engagement recess part **85** (see FIG. **15**) of the floor module **80**. Furthermore, as illustrated in, for example, FIGS. **20** and **21**, the support column engagement part **79b** of the hook **79** is a part bent from the other end of the base part **79c** of the hook **79** to the one end side (lower side) and extending downward, and is formed such that the support column engagement part **79b** can enter the accommodation recess part **75a** of the support column module **70V** and engage with and be fixed to the outer wall part **75f** of the accommodation part **75**. Furthermore, as illustrated in FIGS. **20** and **21**, for example, the base part **79c** of the hook **79** is formed extending along the outer wall part **75f** of the accommodation part **75** of the support column module **70**.

(100) Furthermore, as illustrated in FIG. **20**, for example, the hook **79** can engage with and be fixed to the corresponding outer wall part **75f** of each accommodation recess part **75a** of the accommodation part **75**, i.e., a part of the outer wall part **75f** corresponding to each of the dotted lines **P1** to **P4** (see FIG. **13G**). More specifically, the hook **79** is attachable to the two crossing wall parts at the part of the outer wall part **75f** corresponding to each of the dotted lines **P1** to **P4**.

Furthermore, as illustrated in FIG. 20, a recess part **75h** accommodating the support column engagement part **79b** and the base part **79c** of the hook **79** may be formed on the outer wall part **75f**.

(101) Similar to the support column module **70**, the support column module **70V** can be coupled to the other support column module **70**, and can support the floor module **80** at each of the corner portions **83a** to **83d**. Consequently, the support column module **70V** can also make up the rack **10** similar to the support column module **70**. Note that the support column end module **77** can also be formed like the above-described support column module **70V**. That is, the support column end module **77** may include the hooks **79** instead of the flange part **76a**.

(102) The hook **79** functions similar to the floor reception part **76**, engages the floor module **80** with the engagement recess part **85** at each of the corner portions **83a** to **83d**, and supports the floor module **80**. Thus, the support column module **70V** functions similar to the support column module **70**.

(103) Next, a support column module **90** for forming a passage for movement of the above-described transportation elevator **16** will be described. FIG. 22 is a side view of the support column module **90**, FIGS. 23 and 24 are perspective views of the support column modules **90**, and FIG. 25 is an exploded perspective view of the support column module **90**. Furthermore, FIG. 27 is a perspective view of the transportation elevator **16** and part of a passage of the transportation elevator **16** in the rack **10**. The automated storage and retrieval system **1** or the rack **10** may include the support column modules **90**. Each of the support column modules **90** extends between at least mutually neighboring floors **11**, specifically at least between one floor **11** and another floor **11** located immediately above or beneath the one floor **11**. One support column module **90** is configured to be able to be coupled to another support column module **90**. The support column module **90** differs from the above-described support column module **70** in that a component for movement of the transportation elevator **16** is attached. Furthermore, to attach the component for movement of the transportation elevator **16**, the support column module **90** includes a main body **71A** different from the main body **71**, an insertion plug **72A** different from the insertion plug **72**, and an accommodation plug **73A** different from the accommodation plug **73**. Hereinafter, the components of the support column module **90** which are the same as or have similar functions to those of the support column module **70** will be assigned the same reference numerals, description thereof will be omitted, and the different components will be described.

(104) The component of the support column module **90** for movement of the transportation elevator **16** is, for example, an engagement member **91** which a driving member of the transportation elevator **16** engages with, and is more specifically, for example, a roller chain **91** which one of sprockets **16a** (see FIG. 27) of the transportation elevator **16** engages with as illustrated in FIGS. 22 to 25. The roller chain **91** extends in parallel or substantially parallel to the axis **a1** in the support column module **90**. Furthermore, the roller chain **91** extends between the main body **71A** and the accommodation plug **73A**. Furthermore, the roller chain **91** is provided to the support column module **90** such that the roller chain **91** is located in a space which is the passage of the transportation elevator **16** in the rack **10**. Note that the space which is the passage of the transportation elevator **16** is, for example, a space corresponding to the zone space **S**. Hence, the main body **71A**, the insertion plug **72A**, and the accommodation plug **73A** do not have a part overlapping the space which is the passage of the transportation elevator **16**. As illustrated in, for example, FIG. 25, the main body **71A**, the insertion plug **72A**, and the accommodation plug **73A** form a space **92** extending along the axis **a1**.

(105) More specifically, as illustrated in, for example, FIG. 25, the main body **71A** has no part corresponding to the main body **71** located in the one zone space **S** compared to the main body **71** of the support column module **70**, and does not include, for example, the $\frac{1}{4}$ part of the main body **71** around the axis **a1** corresponding to one of the dotted lines **P1** to **P4** (see FIG. 13G).

Furthermore, as illustrated in, for example, FIG. 25, the insertion plug **72A** has no part

corresponding to the insertion plug **72** located in the one zone space **S** compared to the insertion plug **72**, and does not include, for example, the $\frac{1}{4}$ part of the insertion plug **72** around the axis **a1** corresponding to one of the dotted lines **P1** to **P4** (see FIG. **13G**). Furthermore, as illustrated in, for example, FIG. **25**, the accommodation plug **73A** has no part corresponding to the accommodation plug **73** located in the one zone space **S** compared to the accommodation plug **73**, and does not include, for example, the $\frac{1}{4}$ part of the accommodation plug **73** around the axis **a1** corresponding to one of the dotted lines **P1** to **P4** (see FIG. **13G**). Thus, the support column module **90** has the space **92** formed at, for example, parts of the main body **71A**, the insertion plug **72A**, and the accommodation plug **73A** located in the space which is the passage of the transportation elevator **16**. The space **92** corresponds to, for example, a projection space in the axis **a1** direction at one of the corner portions **83a** to **83d** of the floor module **80**.

(106) As illustrated in, for example, FIGS. **22** to **25**, the roller chain **91** is a chain with ends including a pair of end parts **91a** and **91b**, and extends in the space **92** in the axis **a1** direction. The one end part **91a** of the roller chain **91** is located at a position of the accommodation plug **73A** in the axis **a1** direction in the space **92**, and the one end part **91b** of the roller chain **91** is located at a position of the insertion plug **72A** in the axis **a1** direction in the space **92**. Furthermore, the roller chain **91** is fixed to a fixing member **93** provided in the space **92** and forming a space which can accommodate the roller chain **91**. As illustrated in, for example, FIGS. **22** to **25**, the fixing member **93** is a member whose cross-sectional shape perpendicular to the axis **a1** is a substantially U shape, and includes a fixing member piece **94** which is a member whose cross-sectional shape perpendicular to the axis **a1** is a substantially L shape, and a fixing member piece **95** which is a member fixed to this fixing member piece **94**. The fixing member piece **94** is fixed to, for example, a part of the main body **71A** defining the space **92**, and extends from the position of the insertion plug **72a** to the position of the accommodation plug **73A** in the axis **a1** direction in the space **92** as illustrated in FIGS. **22** to **25**. Furthermore, the fixing member piece **95** is fixed to the fixing member piece **94** facing the fixing member piece **94** in the direction perpendicular to the axis **a1** with a gap extending in the axis **a1** direction interposed therebetween.

(107) As illustrated in FIGS. **22** to **25**, the roller chain **91** includes pins **91d** for coupling each chain piece **91c**. Part of the pins **91d** is/are a pin/pins **91e** whose both ends each protrude from the chain piece **91c**. Although the pin **91d** and the pin **91e** are alternately provided with a fixed interval interposed therebetween in the illustrated examples, and arrangement of the pins **91d** and **91e** is not limited to this. Furthermore, all pins of the roller chain **91** may be the pins **91e**. Note that reference numerals are not assigned to all of the pins **91d** and **91e**, and are omitted in FIG. **25**.

(108) Furthermore, as illustrated in FIG. **25**, recess parts **93a** which can accommodate distal ends of the pins **91d** are formed in the fixing member pieces **94** and **95** so as to meet the pins **91d** of the roller chain **91**. When the pins **91d** are accommodated in the recess parts **93a**, the roller chain **91** is fixed to the fixing member **93** (fixing member pieces **94** and **95**). Note that the components for fixing the roller chain **91** to the fixing member **93** are not limited to these pins **91d** and recess parts **93a**. Components or a method for fixing the roller chain **91** to the fixing member **93** may be other components or methods such as screwing or welding. Note that, similar to the pins **91d** and **91e**, reference numerals are not assigned to all of the recess parts **93a**, and are omitted in FIG. **25**.

(109) Furthermore, as illustrated in FIG. **26**, the roller chain **91** includes clips **96** for coupling to the roller chain **91** of the other support column module **90** to which this support column module **90** is coupled. The clip **96** is a member for coupling the chain piece **91c** of the end part **91a** of the one roller chain **91** and the chain piece **91c** of the end part **91b** of the other roller chain **91**, and can engage with the respective pins **91e** of the two chain pieces **91c** to be coupled. Note that components or a method for coupling the two roller chains **91** are not limited to these clips **96**, and may be other components or methods.

(110) The plurality of support column modules **90** is coupled in the rack **10** similar to the support column modules **70** to form shafts of the transportation elevator **16** extending in the **z** axis

direction. Furthermore, as illustrated in FIG. 27, the four shafts of this transportation elevator **16** are arranged facing each other such that the roller chains **91** extending along the respective shafts face each other so as to form a movement space of the transportation elevator **16**, so that a movement rail and the movement space of the transportation elevator **16** are formed. As described above, the respective roller chains **91** of the plurality of support column modules **90** are coupled by coupling the end part **91a** or the end part **91b** of the one roller chain **91** and the end part **91b** or the end part **91a** of the other roller chain **91** by using the clip **96**. Thus, in the shafts of the transportation elevator **16** formed by coupling the plurality of support column modules **90**, the plurality of roller chains **91** becomes a string of a roller chain.

(111) As illustrated in FIG. 27, in the movement space of the transportation elevator **16**, the four respective sprockets **16a** of the transportation elevator **16** mesh with the roller chains **91** of the four shafts. By driving two electric motors **16c** of the transportation elevator **16**, each sprocket **16a** is rotated via a shaft **16b**, each sprocket **16a** moves on the roller chain **91**, and thereby the transportation elevator **16** moves along the shaft in the z axis direction.

(112) As described above, the rack **10** includes the support column modules **90** forming the movement rails of the transportation elevators **16**, and makes it possible to form the movement rails and the movement spaces of the transportation elevators **16** by combining the support column modules **90**. Furthermore, the support column module **90** has the same or substantially same length (the length in the axis **a1** direction) as that of the support column module **70**. Furthermore, the support column module **90** has part of the floor reception part **76** outside the movement space of the transportation elevator **16**, and can support the floor module **80** outside the movement space of the transportation elevator **16** similar to the support column module **70**. Consequently, by changing to the support column modules **90** the support column modules **70** facing a single zone space S in the rack modules **2** surrounding the one zone space S in an xy axis plane direction, it is possible to form the movement space of the transportation elevator **16** in this one zone space S.

(113) As described above, the four support column modules **90** facing each other form the movement space of the transportation elevator **16** corresponding to the one zone space S, so that the support column modules **90** can make a component unit of the movement space of the transportation elevator **16** smaller. By combining the support column modules **90** with the rack modules **2**, it is possible to easily form the movement spaces of the transportation elevators **16** in the rack **10**. Furthermore, the support column modules **90** can form the movement space of the transportation elevator **16** only in the one zone space S. Consequently, it is possible to form the complicated movement spaces of the transportation elevators **16** in the rack **10**. Furthermore, it is possible to form the movement spaces of the transportation elevators **16** inside the rack **10**, too. Consequently, it is possible to flexibly design the form of the automated storage and retrieval system **1**.

(114) Note that, although the above-described support column modules **90** have the roller chains **91** as the engagement members engaging with the sprockets **16a** of the transportation elevator **16**, the engagement members included in the support column modules **90** are not limited to the roller chains, may be members which the sprockets **16a** can engage with, and may be a rack of a rack and pinion. Furthermore, the support column module **90** may not include the fixing member **93**, and the roller chain **91** may be fixed to the main body **71A**, the insertion plug **72A**, and the accommodation plug **73A**. Furthermore, the support column module **90** may include the hooks **79** instead of the floor reception part **76** similar to the support column module **70V**. Furthermore, the support column module **90** may extend over heights of a plurality of floors, or may extend shorter than the height of one floor. Furthermore, the roller chain **91** may extend longer than the support column module **90** (the main body **71A**, the insertion plug **72A**, and the accommodation plug **73A**), or may be shared between the plurality of support column modules **90**. Furthermore, the support column modules **90** may form a passage of the transportation elevator **16** outside the rack in the automated storage and retrieval system **1**.

(115) This description discloses some embodiments of the subject matter of the present disclosure, and uses examples to enable the one of ordinary skilled in the art to carry out the embodiments of the subject matter of the present disclosure including manufacturing and using an arbitrary device and system and executing an arbitrary incorporated method. The patentable scope of the subject matter of the present disclosure is defined by the claims, and may include other examples achieved by the one of ordinary skilled in the art. These other examples intend to be within the claims in a case where the other examples include components which are not different from wordings of the claims, or in a case where the other examples include equivalent components including non-substantial differences from the wordings of the claims.

Claims

1. An automated storage and retrieval system comprising: a rack storing a plurality of bins for containing items; a transportation robot for transporting the bin in the rack; and a transportation elevator which moves in the rack and can transport the transportation robot, wherein the rack includes a plurality of floors each of which stores each of the plurality of bins, and allows the transportation robot to run along a surface thereof, and the rack includes a plurality of support column modules forming a passage of the transportation elevator, and each of the support column modules extends between at least mutually neighboring floors, and one support column module is configured to be able to be coupled to another support column module each of the support column modules includes an engagement member guiding the transportation elevator, wherein the engagement member of the one support column module is configured to be able to be coupled to the engagement member of the other support column module.
2. The automated storage and retrieval system according to claim 1, wherein the transportation elevator includes at least one driven part to be rotationally driven, and the engagement member is configured to be engaged with the driven part of the transportation elevator.
3. The automated storage and retrieval system according to claim 2, wherein the driven part of the transportation elevator is a sprocket, and the engagement member is a roller chain.
4. The automated storage and retrieval system according to claim 1, wherein each of the support column modules is configured to be able to support the floor adjacent to the passage of the transportation elevator.
5. The automated storage and retrieval system according to claim 4, wherein the engagement member of each of the support column modules is configured to be located in a space of the passage of the transportation elevator, and each of the support column modules is configured to be able to support the floor outside the space of the passage of the transportation elevator.
6. The automated storage and retrieval system according to claim 1, wherein each of the support column modules includes an insertion plug, an accommodation plug, and a main body extending between the insertion plug and the accommodation plug, and the insertion plug and the accommodation plug are configured to be able to be coupled to each other.
7. The automated storage and retrieval system according to claim 6, wherein the main body includes a base part and a plurality of rib parts.
8. The automated storage and retrieval system according to claim 6, wherein the insertion plug and the accommodation plug of the support column module are positioned in opposite ends of the main body and along axis a1 defined along a length of the main body.
9. The automated storage and retrieval system according to claim 6, wherein two support columns being coupled to each other by connecting the insertion plug of one support column to the accommodation plug of the other support column.
10. The automated storage and retrieval system according to claim 6, wherein the insertion plug being defined with a plurality of insertion parts including a plurality of insertion protrusion parts protruding in direction of the axis a1.

11. The automated storage and retrieval system according to claim 10, wherein the insertion protrusion parts being defined with claw parts extending along the axis a1 formed at a distal end of each insertion protrusion parts.
 12. The automated storage and retrieval system according to claim 6, wherein the accommodation plug of a column module includes an accommodation part defined with a plurality of accommodation recess parts configured to accommodate the insertion protrusion parts of the insertion plug of another column module.
 13. The automated storage and retrieval system according to claim 6, wherein the rack includes a floor module including a plurality of corner portions, the corner portions of the floor module configured to be supported by the accommodation plugs of the support column modules.
 14. The automated storage and retrieval system according to claim 13, wherein the accommodation plug comprising: a floor reception part formed around the accommodation part, wherein the floor reception part comprising: a flange part forming a face extending along a plane perpendicular to the axis a1; engagement protrusion parts protruding from the flange part to a side opposite to the main body in the axial direction, wherein the engagement protrusion part being defined with a plurality of claw part and a plurality of pins configured to engage with corresponding parts.
 15. The automated storage and retrieval system according to claim 13, wherein the corner portions of the floor module are defined with cutout faces, each cutout face being defined with an engagement recess part and recess parts configured to accommodate the claw part and the pins of the engagement protrusion part.
 16. The automated storage and retrieval system according to claim 8, wherein the support column module comprising hooks fixed to the accommodation part and configured to engage with the floor module.
 17. The automated storage and retrieval system according to claim 1, wherein the transportation robot being configured to run on a movement passage defined in each floor of the rack.
 18. The automated storage and retrieval system according to claim 1, wherein each of the storage bins face the movement passage defined in each floor of the rack.
 19. The automated storage and retrieval system according to claim 1, wherein the transportation robot is configured to trace a line drawn on the floor of the rack.
 20. The automated storage and retrieval system according to claim 1, wherein each floor of the rack includes one or more charging spots for charging a battery of the transportation robot.
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